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Team Mentor: Gökhan Doğan

Past Experience

- **Erkam Tuna Bayoğlu:** TÜBİTAK 2204-A Tech Design 2nd Place (Turkey) & 2024 FRC Marmara Regional Winner.
- **Ege Serter:** CS & Algorithm expert. Gold Medalist (Istanbul Science Olympics) & 2x Bronze Medalist (TÜBİTAK National Science Olympics).
- **Osman Baran Ayaydın:** Software specialist. 1st & 2nd Place Winner (Turkey) in the TÜBİTAK 2204-A Research Projects Competition.
- **Göktuğ Aslanoğlu:** Computer Vision & Autonomous Systems expert. TÜBİTAK BİLGEM & UME intern; experienced in underwater CV modules and LLMs.

Sensor Strategy

1. **LiDAR:** Mounted on the top center. Used for complex obstacle detection (especially in non-grid Area 4) and high-resolution (360-degree) mapping.
2. **3x RGB Cameras:** Side-mounted cameras simultaneously scan for victims in narrow corridors, while the downward-facing front camera handles floor analysis (time-consuming brown swamps, black holes, and silver checkpoints).
3. **GPS and IMU Integration:** Absolute position (GPS) and rotational heading (IMU Gyroscope) data are fused to ensure flawless SLAM mapping.

Algorithms and Methods

1. Dynamic Navigation (Dijkstra)

- Exploration Planner: Applies a "Turn Penalty" to prevent zig-zagging and ensure smoother trajectories.
- Targeted Planner: Employs a Hysteresis Gate to prevent rotational oscillation in narrow corridors.

2. Advanced Computer Vision (OpenCV)

Uses custom HSV filtering to reject wall reflections and fake 3D targets.

- Cognitive Targets: Analyzes ring thickness via connected-components. Partially merged rings are mathematically extrapolated to 5 based on thickness ratios.
- Letter Victims: Accurately identifies letters by counting white components

3. Grid-Based SLAM Mapping

- Safety Margin (Inflation): Generates a 5mm resolution LiDAR grid. Walls are "inflated" by 2.5cm and holes by 1.2cm to guarantee collision-free paths.
- FOV Tracking: Safe areas within the camera's Field of View (FOV) are instantly marked as "visited" using OpenCV fillPoly. A tilt-guard prevents map corruption on uneven terrain.

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LOOP while simulation runs:
  Read Sensors (LiDAR, GPS, IMU, Cameras)
  IF tilt > 0.01: SKIP iteration (Prevent map error)
  Update Map (Pose, Visited FOV, Obstacles, Holes)

SWITCH robot_state:
  0 (Explore):
    Follow Dijkstra exploration path.
    IF target detected -> State = 1
  1 (Approach Target):
    Navigate to 0.06m from target.
    IF reached -> State = 2
  2 (Align & Identify):
    Rotate to face target.
    IF aligned -> Run Vision algorithms, Mark Done
```

